

Project 2 — Data Classification Using AI

About

Built a Data Classification model using the Iris dataset and Scikit-learn. Loads and explores the dataset, splits it into training and testing sets using `train_test_split`, trains a K-Nearest Neighbors (KNN) classifier, and evaluates the model's accuracy on unseen test data.

Key Concepts Used

- **`load_iris()`**— Loadsthe built-in Iris flower dataset from sklearn
- **`dataset.data` / `dataset.target`** — Separates features (X) and labels (y)
- **`train_test_split()`** — Splits dataset into 70% training and 30% testing sets
- **`KNeighborsClassifier`** — Classifies data based on proximity of K nearest neighbors
- **`.fit()`** — Trains the KNN model on training data
- **`.predict()`** — Generates predictions on the test data
- **`accuracy_score()`** — Measures how accurately the model classifies test samples

Program Output

```
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STEP 1: DATASET OVERVIEW
=====
Keys      : dict_keys(['data', 'target', 'frame', 'target_names', 'DESCR', 'feature_names', 'filename',
'data_module'])
Number of rows: 150
Number of cols: 4
Shape     : (150, 4)
Feature Names : ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']

Top 5 rows:
[[5.1 3.5 1.4 0.2]
 [4.9 3.  1.4 0.2]
 [4.7 3.2 1.3 0.2]
 [4.6 3.1 1.5 0.2]
 [5.  3.6 1.4 0.2]]
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STEP 2: TRAIN-TEST SPLIT
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X_train shape: (105, 4)
y_train shape: (105,)
X_test shape : (45, 4)
y_test shape : (45,)
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STEP 3: MODEL EVALUATION
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Accuracy: 1.00
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```